Disadvantages

- Preparation of screen is a time consuming process
- Preservation of screen needs extra care
- A small damage in the screen will spoil the entire printing

The design is created by painting or making non-design portions of the screen opaque, thus preventing the print paste from passing through. The areas where the print paste passes through will create a printed pattern. The screen is placed in contact with the fabric to be printed and the print paste is forced through the screen by a squeegee. The squeegee is used to spread the dye evenly through the screen. It is moved across the screen, forcing the print paste through the mesh openings. It helps in making a clean image on the printed surface. A screen is prepared for each colour of the design. There are two types of screen printing namely Flat screen printing and Rotary screen printing.

Flat Screen Printing or Hand Screen Printing

It is done by hand. The design is copied onto a series of very fine, flat screens, one for each colour to be printed. Lacquer or other impermeable substance is applied to all parts of the screen that are not part of its design. Each screen is fitted onto a wooden or metal frame. The fabric to be printed is spread onto a long table. A screen is set over the fabric and the printing paste is poured on the screen and forced through its unblocked areas onto the fabric with a squeegee. The screen is then moved to the next section of the fabric and the operation is repeated until the entire fabric is printed. This process is repeated for each colour of the design. Hand screen printing is time-consuming and limited to short length of fabrics.

Automatic Flat Screen Printing

In this method, the process is automated and therefore faster. Here the fabric moves on a wide rubberized belt. The screens are placed above the belt. As the fabric moves, the screens are automatically lowered to the cloth and the appropriate colour is applied with automatically regulated squeegees. The cloth is dried in an oven.

Advantages

- Prints upto twenty colours in one run
- High production rate
- Produce brighter and cleaner shades
- Produces designs consisting of squares, circles and ovals

Disadvantage

- High cost

Rotary Screen Printing

This method of printing is done using machine. The fabric to be printed is moved on a wide rubber belt under the rotary screen cylinders. It is the fastest method of screen printing, with a production of more than 3500 yards per hour. A squeegee in each rotary screen forces the paste through the screen onto the fabric. The cloth is then passed into a drying oven to set the colour and washed.

Advantages

- Faster method of printing
- Rotary metal screens are light weight in contrast to the heavy copper rollers and hence they cost less.

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